ENGINEERING BLUEPRINT / VERSION 0.7

Team Unicorn — Technical & Simulation Status

Whole number-system universe construction / release 0.7

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The Whole Number-System Universe Blueprint

The overarching objective is an engineering realisation of an already-defined abstract number system. The integer stepping-stone line is its foundation—not the completed numerical universe.

Layer	Blueprint wording	Implementation meaning
The Blueprint — Abstract Number System	Integers, rationals, reals, operations and relations	The logical specification of what is to be represented
Phase I — Integer Foundation	+1 marking; unbounded integer stepping stones	Successor markers, correct labels, unlimited extension
Phase II — Structural Engineering	Prime-factor checksum; links, reuse, rearrangement	Cumulative certificates, layout and management
Phase III — Numerical Continuum	Rational and real representations between integers	Lazy interval access; division and completeness
Phase IV — Dynamic Mathematical Universe	3D construction, time, Fourier, morphogenesis	Dynamical spatial implementation and propagation

Sequence of layers versus concurrent construction

These phases specify logical layers, not four strictly serial milestones. Successor marking (I), inherited structural certification (II), and time-dependent gait/feedback (IV) operate concurrently from the outset. The numerical continuum (III) needs separately specified representations and completeness rules before it can be implemented.

Construction process: Team Unicorn Relay

Operation	Assignment	Acceptance criterion
설계 / **Design**	Pre-known destination n; choose leg count, candidate prime leg labels, phase, cadence and manufacturing cost	No hard-coded prime success rule
이동·인증 / **Transit**	Travel across and audit the previously certified prefix 0..n−1 using inherited certificates	0/1 exceptions; cache invariants
+1 마킹 / Marking	Final successor +1 from n−1 to n	Exactly one new marker
실수 구간 / **Interval**	Mark interval [n−1,n] as potentially addressable	Promise of representation, not creation of reals

귀환·개선 / **Handover**	Return performance, audit, control and layout information for next generation	Include return and redesign cost

Spatial and informational stack

Layer	Role
1D number line	The visible integer markers and exact successor order
Prime-factor graph	Inherited certificates, multiplicative relationships and caches
3D placement	Cubic placement and spherical expansion as testable hybrid components
4D dynamics	Separate elapsed time τ from generation index t; Fourier and feedback
HQ memory	Inherited state, design updates, time/coordination/entropy cost

Non-negotiable mathematical constraints

- The build implements representations of mathematical numbers; it does not create their mathematical existence.
- Infinity means no fixed finite extension limit, not completion of infinitely many individual markers in finite time.
- Integer markers alone do not implement the real continuum: representation and completion rules are additional.
- The leg count k need not be prime; prime-labelled individual legs do not imply stride equals prime value.
- Report wall-time, operation count, communication, re-layout, coordination and validation separately.

Next engineering milestones

- M1 Implement successor marking with genuinely inherited arithmetic certificates.
- M2 Introduce in-flight feedback using distance remaining to a known target.
- M3 Compare prime/composite/shuffled leg labels and 1–10-second vs 1–1M-target experiments.
- M4 Compare cubic/spherical/hybrid layouts under identical certified workloads.
- M5 Compare physical return, information-only relay and distributed HQ, including generational learning.

Research programme retained from v0.6

0. Document purpose

This is the technical companion to Research Proposal v0.6. It consolidates model assumptions, actual recorded computations and outstanding gaps. None of the simulations is empirical evidence for a physical cosmos or a new prime law. The task remains building integer stepping stones, not searching for primes.

1. Experiment lineage

Version	Coverage	Assumption or purpose
0.2	1 / 5 / 10 seconds	Hypothetical distance × cadence, coordination and queueing
0.3	n=2..100	Known targets, prime-labelled leg = stride as a toy assumption
0.4	n=2..100	No prime stride; rectified sine propulsion and separate Fourier divisibility
0.5	n=2..1,000,000	Same 0.4 rules scaled to one million targets
0.6	1 / 5 / 10 seconds	Phase disturbances and local in-flight feedback
0.6.1	1 / 5 / 10 seconds	Independent coordination cost and feedback control price

2. Gait Lab 0.3: destination-aware trial

The known target n determines each journey origin → n−1 → mandatory final +1 to n, with optional return. The solver compared 209 multiset architectures (1–4 legs, primes 2, 3, 5, 7, 11 and 13, with repetition).

Crucial assumption: each active leg contributes its numerical prime label directly to forward displacement in each pulse. This is not a derived leg physics. A separate fine +1 adjustment was also allowed during transit.

Under the illustrative medium-coordination, inherited-catalogue, round-trip condition, the best architecture had one leg for 19/99 targets and two for 80/99. Aggregate time improved by about 29.6% versus the best permitted one-leg design for each target. Preloaded versus inherited prime catalogues were compared.

3. Fourier Lab 0.4: no assumed prime strides

The candidate catalogue contains 2,924 architectures with 1–3 legs, frequencies 1–6 Hz and four initial phase offsets. An activation $a_i(\tau)=\max(0,\sin(2\pi f_i\tau+\varphi_i))$ feeds a hypothetical rectified propulsion law.

Among 99 destinations the default model selected one leg 12 times, two legs 87 times and three legs zero times. Most selected active frequencies were 1Hz. Since primality and prime labels never enter the propulsion function, no prime-specific gait can be inferred.

A separate roots-of-unity identity $\delta_q(n)=(1/q)\Sigma_r \exp(2\pi irn/q)$ reproduces whether q divides n. It is exact standard Fourier arithmetic and can represent primality checking over $q\leq\sqrt{n}$, but does not explain how locomotion would generate arithmetic structure.

4. Scale Lab 0.5: results to one million

The 0.4 propulsion and cost laws were unchanged across destinations 2…1,000,000 (999,999 cases), with 2,924 candidate gaits per destination. The optimised search was cross-checked by 409 exhaustive targets, including the original 99.

Default winner counts: one leg 12, two legs 999,987, three legs zero. At $n\geq40$ only two opposite-phase 1Hz two-leg gaits won. Six distinct winners appeared in total. Total model time was 21.51% below the sum of destination-specific one-leg baselines.

Under a strictly serial origin-roundtrip relay, the total is approximately 246,583,683,620 model seconds, equivalent to 7,814 notional model-years at ordinary unit conversion. This is not empirical physical time; it exposes the cumulative distance bottleneck.

Prime/composite status was computed separately and was not used by the optimiser. No prime-specific transition emerged. Changing management cost can make one, two or three legs optimal at long range.

5. Feedback 0.6 and coordination 0.6.1

Feedback 0.6 added phase kicks at 0.75, 3 and 7 seconds and corrective angular velocities. In one illustrative 10-second two-leg comparison, fixed gait travelled 43.32 units, adaptive gait 44.92 units and undisturbed gait 45.01 units. This is phase-recovery in a synthetic controller, not prime-related motion.

Lab 0.6.1 separately prices pairwise coordination η and feedback-control effort. The coordination charge includes connected leg-pair time and accumulated phase mismatch. It changes the adjusted score but does not itself physically reduce the propulsion speed.

Neither current feedback lab implements feedback from the known remaining destination, inherited full integer certificates, 3D spatial rearrangement, homeward optimisation or learning of the HQ design rule across unicorn generations.

6. Revised model and missing components

Component	Implemented now	Required next

7. Next experimental controls

- A — Fixed versus adaptive destination-aware feedback using equal propulsion rules.
- B — Prime, composite and shuffled leg labels under identical mechanics.
- C — Re-factor-the-whole-prefix audit versus inherited certificate verification.
- D — Physical return versus information-only handover versus distributed HQ.
- E — Cube, sphere and hybrid under equal arithmetic workload, relayout and communication costs.
- F — Hold-out destinations and out-of-sample rules to catch overfitting.

8. What the results do and do not establish

These runs have produced a reproducible experimental framework for testing which assumptions affect time, coordination, feedback and architecture selection. The models have not demonstrated that nature uses primes as legs, that a Fourier gait generates the primes or that the universe is optimally constructed by unicorn agents.

Claims should be consistently labelled as mathematical fact, code-derived simulated output, testable hypothesis or metaphor.

9. Provenance and versions

Scope comes from the v0.4–v0.5 proposals and explicit user corrections through 13 October 2026. Computation sources: Gait Lab 0.2 archive, 0.3 source archive, Fourier 0.4 research archive, Scale 0.5 archive and 1M CSV, Feedback 0.6 archive and Feedback 0.6.1 HTML.

This document records source-reported numerical outcomes and does not claim a fresh independent proof or an exhaustive new literature review.